

Instanton interactions at the one-loop level

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We discuss the interactions between widely separated instantons (but *not* instantons and anti-instantons) due to the one-loop fluctuations in a non-Abelian gauge theory. A slightly generalized version of the recent calculation of Brown and Creamer for the one-loop determinant about an exact self-dual solution is used as a starting point. An apparently dangerous $1/r^2$ interaction is found to disappear with the proper parametrization of the exact solution. As a by-product of our procedure, we develop a more precise understanding of the relation between exact solutions of topological charge k and the superposition of k instantons. In particular, the meaning of the parameters of the Jackiw-Nohl-Rebbi $k = 2$ solution is understood in these terms.

I. INTRODUCTION

Pseudoparticles, self-dual or anti-self-dual solutions¹ to the classical Yang-Mills equation in Euclidean space, are believed to play an important role in determining the physical predictions of a non-Abelian gauge theory. Unfortunately, calculations of the consequences of these solutions²⁻⁵ have usually proved tractable only in the so-called dilute-gas limit where the size λ of the pseudoparticles is much smaller than the typical spacing r between them. In this limit, the leading quantum effects of the many-pseudoparticle sector are calculated by allowing the quantum fields to interact independently with each pseudoparticle.

One might wish to investigate the corrections to the leading term of this approximation by calculating the one-loop action about superpositions of instantons (self-dual solutions) and anti-instantons (anti-self-dual solutions). One way that has been suggested⁴ to accomplish this would be to introduce constraints in such a way as to make the superposition an *exact* solution, and then integrate over the constraints. However, this procedure seems difficult to implement rigorously. Here we discuss a much simpler problem, the one-loop corrections to a superposition of instantons alone (no anti-instantons). Such superpositions have the useful property that they are very close to *exact* solutions. This fact enables us to avoid the difficulties of quantum mechanics about approximate solutions; we detour through the exact solution and then reinterpret the results in terms of the contributions of individual instantons. Note that we explicitly discuss only a finite number of widely separated instantons; we will have nothing to say about the subtle problems of an infinite number of instantons (the true dilute-gas limit of small but finite density and infinite volume). Further, since there exists no exact solu-

tion which corresponds to an unconstrained superposition of instantons and anti-instantons, the physically most interesting situation is inaccessible to our treatment. Still, one expects that the size of the interactions between widely separated instantons provides some kind of lower bound to the size of interactions between widely separated instantons and anti-instantons.

In order to carry out our program it is first necessary to establish precise connections, at the classical level, between exact solutions and linear superpositions of instantons. It is by now well accepted that the $8k-3$ parameters⁶ of an $SU(2)$ solution of topological charge k can be interpreted as describing the positions, scales, and relative gauge orientations of k instantons. A corresponding interpretation exists in an arbitrary gauge group.⁷ We show to what extent this interpretation makes sense. We study in particular the topological charge-two solution written down by Jackiw, Nohl, and Rebbi⁹ (JNR) as a generalization of the solutions discovered by 't Hooft.⁸ Because this JNR solution is known to contain all 13 parameters necessary to describe two instantons, it provides a simple testing ground for many of our ideas. We gain a detailed understanding of the meaning of the JNR parameters in the limit of large separation.

In order to study quantum corrections, we generalize slightly the recent calculation of Brown and Creamer.¹⁰ These authors compute the one-loop action about the 't Hooft solutions; we extend their result to the JNR solutions. We then examine quantum fluctuations about two distant, arbitrarily oriented instantons and show that the one-loop corrections to the two-instanton action are smaller than λ^2/r^2 . Corrections to the collective-coordinate Jacobian are shown to have the same bound. Note that if a λ^2/r^2 correction appeared, it would represent a strong Coulomb in-

teraction (in 4 dimensions) which might cause a total breakdown of the dilute-gas approximation by introducing long-range correlations between instantons. (Such an interaction, of course, loses this significance if there are light fermions of mass $\ll \lambda^{-1}$, which already destroy the dilute-gas approximation by producing logarithmic forces between instantons and anti-instantons.⁴⁾ The understanding of the parameters of the JNR solution proves crucial to our calculation. Similar results can be obtained for arbitrary instanton number if all the instantons are aligned ('t Hooft solution).

The outline of this paper is as follows: In Sec. II we discuss some general properties of a classical superposition of instantons. In particular, we estimate the amount by which the superposed potentials differ from the potential of the appropriate exact solution. Section III examines the JNR $k=2$ solution in detail, interpreting the parameters of this solution in terms of the positions, scales, and relative orientation of the corresponding superposition. Sections II and III discuss classical properties of the solutions exclusively and are completely self-contained. In Sec. IV, we study the interactions between two instantons due to quantum corrections. We do this by expanding the one-loop action in the widely separated limit. The collective-coordinate Jacobian is also discussed, and a subtle point about the meaning of pseudoparticle interactions is explained. Section V contains a summary of our results and some additional remarks. There are two appendixes: Appendix A discusses some additional details of the JNR $k=2$ solution; Appendix B generalizes the results of Ref. 10 to the JNR solutions.

II. GENERAL CLASSICAL PROPERTIES OF THE SUPERPOSITION

For the purposes of this section, we may consider a Yang-Mills theory in an arbitrary simple gauge group G . The gauge potential and field strength are

$$A_\mu = A_\mu^a T^a, \quad (2.1a)$$

$$F_{\mu\nu} = F_{\mu\nu}^a T^a = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu], \quad (2.1b)$$

where T^a are the group generators. The Euclidean space action and topological charge are then given by

$$S = \alpha \int d^4x \mathcal{S}, \quad \mathcal{S} \equiv \text{tr} F_{\mu\nu} F_{\mu\nu}, \quad (2.2a)$$

$$\mathcal{S} = \sum_i \text{tr}(F_{\mu\nu}^{(i)})^2 + \sum_{i \neq j} \text{tr} F_{\mu\nu}^{(i)} F_{\mu\nu}^{(j)} - 4i \sum_{i \neq j} \text{tr} F_{\mu\nu}^{(i)} [A_\mu^{(i)}, A_\nu^{(j)}] - 2i \sum_{i \neq j \neq k} \text{tr} F_{\mu\nu}^{(k)} [A_\mu^{(i)}, A_\nu^{(j)}] - \text{tr} \left(\sum_{i \neq j} [A_\mu^{(i)}, A_\nu^{(j)}] \right)^2. \quad (2.9)$$

$$k = \beta \int d^4x \text{tr} F_{\mu\nu} F_{\lambda\sigma} \epsilon_{\mu\nu\lambda\sigma}, \quad (2.2b)$$

where α and β are constants which depend on the normalization of the generators. The equation of motion is

$$D_\mu F_{\mu\nu} \equiv \partial_\mu F_{\mu\nu} - i[A_\mu, F_{\mu\nu}] = 0. \quad (2.3)$$

If J^i , $1 \leq i \leq 3$, generate a minimal⁷ SU(2) subgroup of G , normalized by

$$[J^i, J^j] = i\epsilon^{ijk} J^k, \quad (2.4)$$

then a singular gauge instanton with topological charge one, scale λ , position y , and group orientation $g \in G$ can be written as

$$A_\mu(x) = g^{-1} (-J^k \bar{\eta}_{k\mu\nu} \partial_\nu \ln \Pi) g, \quad (2.5)$$

$$\Pi(x) = 1 + \frac{\lambda^2}{(x-y)^2},$$

where $\bar{\eta}_{k\mu\nu}$ is defined by 't Hooft⁴: $\bar{\eta}_{k\mu\nu} = \epsilon_{k\mu\nu}$ if $\mu, \nu = 1, 2, 3$, $\bar{\eta}_{k4\mu} = \delta_{k\mu}$, $\bar{\eta}_{k\mu 4} = -\delta_{k\mu}$, $\bar{\eta}_{k44} = 0$. Recall that A_μ has the following behavior:

$$|A_\mu| \sim \frac{\lambda^2}{|x-y|^3} \quad \text{for } |x-y| \gg \lambda, \quad (2.6a)$$

$$|A_\mu| \sim \frac{1}{|x-y|} \quad \text{for } |x-y| \ll \lambda. \quad (2.6b)$$

The singularity for small $|x-y|$ is, of course, a gauge artifact and does not appear in the action density $\mathcal{S}$.

We are interested in comparing the linear superposition of arbitrarily oriented instantons with an exact solution of the same topological charge. In the case where the typical instanton size λ is much less than the typical instanton separation r , one expects that the superposition will be "close" to an exact solution. We thus write

$$\sum_i A_\mu^{(i)}(x) = A_\mu^E(x) + \Theta_\mu(x), \quad (2.7)$$

where $A_\mu^{(i)}(x)$ is the gauge field due to the i th singular gauge instanton with orientation $g^{(i)}$, A_μ^E is an appropriately chosen exact solution, and $\Theta_\mu(x)$ is a correction term whose size we wish to estimate. The choice of A_μ^E which makes Θ_μ as small as possible will become obvious shortly.

We will estimate Θ_μ by calculating the action of both sides of Eq. (2.7). The field strength of the superposition $A_\mu = \sum_i A_\mu^{(i)}(x)$ is

$$F_{\mu\nu} = \sum_i F_{\mu\nu}^{(i)} - i \sum_{i \neq j} [A_\mu^{(i)}, A_\nu^{(j)}], \quad (2.8)$$

and the action density is

Now note that for $i \neq j$

$$\begin{aligned} \int d^4x \operatorname{tr} F_{\mu\nu}^{(i)} (F_{\mu\nu}^{(j)} - 2i[A_\mu^{(i)}, A_\nu^{(j)}]) &= \int d^4x \operatorname{tr} F_{\mu\nu}^{(i)} (2\partial_\mu A_\nu^{(j)} - 2i[A_\mu^{(i)}, A_\nu^{(j)}] - i[A_\mu^{(j)}, A_\nu^{(i)}]) \\ &= -i \int d^4x \operatorname{tr} F_{\mu\nu}^{(i)} [A_\mu^{(j)}, A_\nu^{(j)}], \end{aligned} \quad (2.10)$$

where we have integrated by parts (the singularities are sufficiently weak to allow this) and have used $D_\mu^{(i)} F_{\mu\nu}^{(i)} = 0$. Using Eq. (2.10) and defining the “Abelian” part of $F_{\mu\nu}^{(i)}$ as

$$\mathfrak{F}_{\mu\nu}^{(i)} \equiv \partial_\mu A_\nu^{(i)} - \partial_\nu A_\mu^{(i)} \quad (2.11)$$

allows us to write

$$\begin{aligned} \frac{\Delta S}{\alpha} &\equiv \int \mathfrak{s} - \sum_i \int \operatorname{tr} (F_{\mu\nu}^{(i)})^2 = - \sum_{i \neq j} \int \operatorname{tr} \mathfrak{F}_{\mu\nu}^{(i)} \mathfrak{F}_{\mu\nu}^{(j)} + \sum_{i \neq j} \operatorname{tr} [A_\mu^{(i)}, A_\nu^{(i)}] [A_\mu^{(j)}, A_\nu^{(j)}] \\ &\quad - 2i \sum_{i \neq j \neq k} \int \operatorname{tr} F_{\mu\nu}^{(k)} [A_\mu^{(i)}, A_\nu^{(j)}] - \int \operatorname{tr} \left(\sum_{i \neq j} [A_\mu^{(i)}, A_\nu^{(j)}] \right)^2. \end{aligned} \quad (2.12)$$

The last three terms on the right-hand side of Eq. (2.12) are all of order λ^6/r^6 . This can be seen by considering separately the two important regions¹¹ contributing to the integrals of these terms: (a) near any instanton ($|x - y_i| \lesssim \lambda$ for some y_i) and (b) far from all instantons ($|x - y_i| \gtrsim r$ for all y_i). For contributions of type (a) the magnitude of the integrand is never greater than λ^2/r^6 , since it always involves at least two potentials from other instantons—each of which falls like $1/r^3$. The volume of integration is of order λ^4 , so region (a) contributes in order λ^6/r^6 . In region (b) the integrand is at most λ^6/r^{10} (from two potentials and one field strength). The volume of integration is of order r^4 , so region (b) again contributes in order λ^6/r^6 . We thus have

$$\frac{\Delta S}{\alpha} = - \sum_{i \neq j} \int \operatorname{tr} \mathfrak{F}_{\mu\nu}^{(i)} \mathfrak{F}_{\mu\nu}^{(j)} + O\left(\frac{\lambda^6}{r^6}\right). \quad (2.13)$$

The right-hand side of Eq. (2.13) is ostensibly of order λ^4/r^4 , as one can see by arguments paralleling the ones above. However, use of the precise form of the instanton solution [Eq. (2.5)] results in additional cancellations. After an integration by parts, Eq. (2.13) becomes

$$\frac{\Delta S}{\alpha} = 2 \sum_{i \neq j} \int d^4x \operatorname{tr} (\partial_\mu \mathfrak{F}_{\mu\nu}^{(i)} A_\nu^{(j)}) + O\left(\frac{\lambda^6}{r^6}\right). \quad (2.14)$$

Using Eq. (2.5) and the fact that $\square\Pi/\Pi = 0$, we have

$$\partial_\mu \mathfrak{F}_{\mu\nu}^{(i)} = g_{(i)}^{-1} \left[J^k \bar{\eta}_{k\nu\alpha} \partial_\alpha \left(\frac{[\partial_\mu \Pi_{(i)}]^2}{\Pi_{(i)}} \right) \right] g_{(i)}. \quad (2.15)$$

Substituting Eq. (2.15) into Eq. (2.14), integrating by parts again, and using the identities $\bar{\eta}_{k\nu\alpha} \bar{\eta}_{k'\nu\beta} = \delta_{kk'} \delta_{\alpha\beta} + \epsilon_{kk'm} \bar{\eta}_{m\alpha\beta}$ and $\square\Pi/\Pi = 0$ then gives

$$\begin{aligned} \frac{\Delta S}{\alpha} &= 2 \sum_{i \neq j} \int d^4x \operatorname{tr} (g_{(i)}^{-1} J^k g_{(i)} g_{(j)}^{-1} J^k g_{(j)}) \\ &\quad \times \frac{(\partial_\mu \Pi_{(i)})^2}{\Pi_{(i)}} \frac{(\partial_\nu \Pi_{(j)})^2}{\Pi_{(j)}} + O\left(\frac{\lambda^6}{r^6}\right). \end{aligned} \quad (2.16)$$

Now, since $(\partial_\mu \Pi_{(i)})^2$ falls like $|x - y_i|^{-6}$ at large distances, the integrand is of order λ^2/r^6 in region (a) and of order λ^8/r^{12} in region (b), so the integral is of order λ^6/r^6 . We thus have

$$\Delta S = O\left(\frac{\lambda^6}{r^6}\right). \quad (2.17)$$

Note that this result depends on the fact that we have superposed instantons only; if we add instantons and anti-instantons, ΔS will be of order λ^4/r^4 . Presumably, ΔS is larger in this case because there exists no exact solution for finite λ/r which “looks like” a superposition of instantons and anti-instantons. We note that these results for ΔS are not new; they were obtained in Ref. 4 in a somewhat different setting.

We now calculate ΔS from the right-hand side of Eq. (2.7) and use the result to estimate Θ_μ . The action of the potential $A_\mu = A_\mu^E + \Theta_\mu$ is given by

$$\begin{aligned} \frac{S}{\alpha} &= \int \operatorname{tr} (F_{\mu\nu}^E)^2 - 2 \int \operatorname{tr} \Theta_\mu \mathfrak{D}_{\mu\nu} \Theta_\nu \\ &\quad + \text{higher order in } \Theta_\mu, \end{aligned} \quad (2.18)$$

where

$$\mathfrak{D}_{\mu\nu} \equiv (D^E)^2 \delta_{\mu\nu} - D_\mu^E D_\nu^E - 4i F_{\mu\nu}^E. \quad (2.19)$$

Of course, terms linear in Θ_μ do not appear in Eq. (2.18) because A_μ^E is an exact solution, that is,

$$\left. \frac{\delta S}{\delta A_\mu} \right|_{A_\mu = A_\mu^E} = 0. \quad (2.20)$$

Since we have chosen A_μ^E to be a self-dual solution with the same topological charge as the superposition, its action is just the sum of the actions of individual instantons. Thus Eqs. (2.17) and (2.18) and the assumption that Θ_μ is small give

$$\int \text{tr} \Theta_\mu \mathfrak{D}_{\mu\nu} \Theta_\nu = O\left(\frac{\lambda^6}{r^6}\right). \quad (2.21)$$

We now require that Θ_ν contain no zero modes of the operator $\mathfrak{D}_{\mu\nu}$. If zero modes are present, we simply absorb them into A_μ^E , creating a new exact solution with slightly shifted parameters. Since Θ_ν has no piece that is annihilated by $\mathfrak{D}_{\mu\nu}$, its magnitude may be estimated from Eq. (2.21). Near any instanton, $\mathfrak{D}_{\mu\nu}$ has magnitude $1/\lambda^2$ (from two derivatives, a derivative and a potential, or two potentials): far from all instantons, $\mathfrak{D}_{\mu\nu}$ has magnitude $1/r^2$ (from two derivatives). Requiring that the left-hand side of Eq. (2.21) be no larger than $O(\lambda^6/r^6)$ then implies

$$\Theta_\mu \lesssim \begin{cases} \lambda^2/r^3 & \text{near region } (|x-y_i| \lesssim \lambda \text{ for some } y_i), \\ \lambda^3/r^4 & \text{far region } (|x-y_i| \gtrsim r \text{ for all } y_i). \end{cases} \quad (2.22)$$

Note that Θ_μ is required to saturate at least one of these bounds, since the contribution to Eq. (2.21) must come from somewhere. This behavior is confirmed by the detailed analysis of the JNR $k=2$ solution which appears in the following section.

III. INTERPRETATION OF THE PARAMETERS IN THE $k=2$ SOLUTION

We now turn to an examination of the general two-instanton solution in $SU(2)$ obtained by Jackiw, Nohl, and Rebbi.⁹ Although the meaning of the parameters is somewhat more accessible in the new solutions of Atiyah *et al.*,^{12,13} the one-loop action¹⁰ has a very simple form in terms of the harmonic pseudopotential Π that appears in the JNR solution, and we will therefore work with the latter.

The JNR expression for the general $k=2$ solution in $SU(2)$ is

$$A_\mu^{\text{JNR}}(x) = -\bar{\sigma}_{\mu\nu} \partial_\nu \ln \Pi_{\text{JNR}}, \quad (3.1)$$

where we have defined $\bar{\sigma}_{\mu\nu} \equiv (\sigma_a/2)\bar{\eta}_{a\mu\nu}$, with σ_a the Pauli matrices, and where

$$\Pi_{\text{JNR}}(x) = \frac{l^2}{(x-z)^2} + \frac{\lambda_1^2}{(x-y_1)^2} + \frac{\lambda_2^2}{(x-y_2)^2}. \quad (3.2)$$

The notation of Eq. (3.2) singles out one of the three points for reasons which will become clear presently.

The first problem in interpreting the param-

eters of Eq. (3.2) is that there are 14 of them ($l, \lambda_1, \lambda_2, z_\mu, y_{1\mu}, y_{2\mu}$, with an overall scale in l, λ_1, λ_2 , clearly irrelevant), whereas it is known⁶ that the most general $k=2$ solution contains only 13 parameters. JNR show that an infinitesimal translation of the positions z, y_1, y_2 around the circle joining them (along with certain changes in the scales) is just a pure gauge transformation. In principle, this transformation may be iterated to obtain a one-parameter family of equivalent configurations, all of whose points lie on the same circle. (In Appendix A, we check this conclusion by exhibiting two equivalent configurations which differ from one another by a finite translation around the circle.) Here we will not need the specific form of the finite transformation; instead we will choose a representative from each family by requiring that the point z lie on the hyperplane which is the perpendicular bisector of the line joining y_1 and y_2 . Such a choice is certainly possible in the dilute-gas limit, as the methods of this section will show. (Of course, because the circle intersects the hyperplane in *two* places, there is still a discrete symmetry among the parameters which does not change gauge-invariant quantities. The meaning of the discrete symmetry is explained in Appendix A.) Finally, let us fix the overall scale of l, λ_1 , and λ_2 , by requiring

$$\begin{aligned} l^2 &= l_1^2 = l_2^2, \\ l_i &\equiv y_i - z, \end{aligned} \quad (3.3)$$

which is made possible by the chosen location of z . We now have a 13-parameter solution: $\lambda_1, \lambda_2, y_{1\mu}, y_{2\mu}$, and 3 components of z . The situation is pictured in Fig. 1.

We now claim that the widely separated limit is given by

$$\lambda_1^2 \sim \lambda_2^2 \sim \lambda^2 \ll r^2 \equiv (y_1 - y_2)^2. \quad (3.4)$$

Furthermore, it is claimed that in this limit, the suitably gauge-transformed solution “looks like” the superposition of two instantons with scales close to λ_1, λ_2 , positions close to y_1, y_2 , and relative gauge orientation determined by the location of z . The claim is not new; to zeroth order [$O(1/\lambda)$ in the near region, $O(\lambda/r^2)$ in the far region] it already appears in the literature.^{7,14} However, here we need to make the correspondence more precise by including terms up to the size of Θ_μ in Eq. (2.22), which is the order in which a superposition stops behaving like an exact solution. This will determine a new expression for the solution which uses as parameters the positions, scales, and group orientation of the superposition which it most closely resembles. The new parametrization will be important for

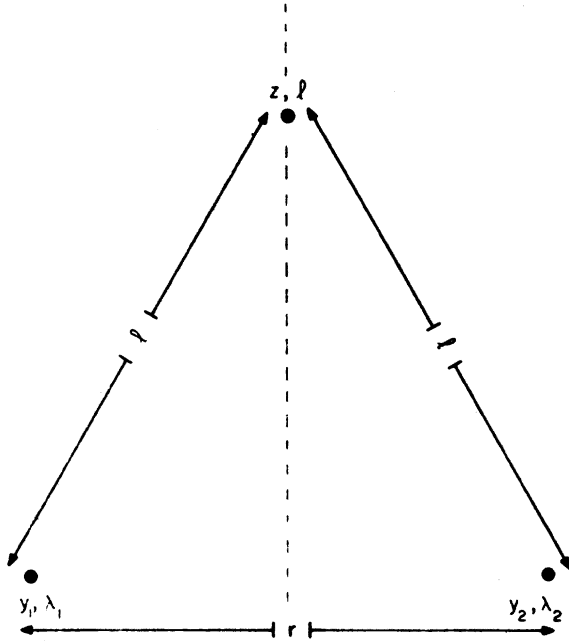


FIG. 1. The location of the points and the corresponding scales of the JNR $k=2$ solution in the widely separated limit ($\lambda_1, \lambda_2 \ll r$). We have used the freedom inherent in the solution to choose z on the hyperplane (dashed line) which perpendicularly bisects the line between y_1 and y_2 , and to choose the scale l corresponding to the point z such that $l = |y_1 - z| = |y_2 - z|$.

describing the quantum corrections to the solution.

The major problem in comparing the JNR solution to a superposition is the location of the gauge singularities. In order to write an equation such as (2.7), with Θ_μ small, it is necessary to make gauge transformations so that both sides of the equation have singularities in the same places. Rather than get involved at once in this tricky problem, let us examine first a gauge-invariant quantity, the action density $\mathcal{S}$. For the JNR solution, Eq. (3.1), and for a single instanton, Eq. (2.5) specialized to $SU(2)$, $\mathcal{S}$ is given by⁹

$$\mathcal{S} = \square \square \ln \Pi, \quad (3.5)$$

where we are to ignore all δ -function singularities in Π , i.e., take

$$\square \square \ln x^2 = 0. \quad (3.6)$$

Let us examine $\mathcal{S}_{\text{JNR}}$ in the neighborhood of y_1 ($|x - y_1| \lesssim \lambda$). Equations (3.2), (3.3), and (3.6) give

$$\mathcal{S}_{\text{JNR}} = \square \square \ln \left[\xi^2 + \lambda_1^2 \left(\frac{\xi^2}{l^2} + \frac{2\xi \cdot l_1}{l^2} + 1 \right) + \frac{\lambda_2^2 \xi^2}{r^2} \frac{(1 + 2\xi \cdot l_1/l^2 + \xi^2/l^2)}{(1 + 2r \cdot \xi/r^2 + \xi^2/r^2)} \right] \quad (3.7)$$

with the definition

$$(x - y_1)_\mu \equiv \xi_\mu. \quad (3.8)$$

For $|\xi| \lesssim \lambda$ Eq. (3.7) becomes

$$\mathcal{S}_{\text{JNR}}(\xi) = \square \square \ln \left[\xi^2 \left(1 + \frac{\lambda_1^2}{l^2} + \frac{\lambda_2^2}{r^2} \right) + \frac{2\lambda_1^2 \xi \cdot l_1}{l^2} + \lambda_1^2 \right] + O\left(\frac{1}{\lambda r^3}\right), \quad (3.9)$$

where we have made use of the fact that l^2 is always greater than, or on the order of, r^2 (see Fig. 1). Now compare Eq. (3.9) to the action density of a superposition of two instantons, one of which is located at position $\hat{y}_1 \equiv y_1 + \zeta_1$ and has scale ρ_1 and the other of which has scale $\sim \lambda$ and is located a distance $\sim r$ away:

$$\mathcal{S}_s(\xi) = \square \square \ln [(\xi - \zeta_1)^2 + \rho_1^2] + O\left(\frac{1}{\lambda r^3}\right), \quad \text{for } |\xi| \lesssim \lambda \quad (3.10)$$

where the term $O(1/\lambda r^3)$ is just the lowest-order effect of the distant instanton. An overall scale in the argument of the logarithm is obviously irrelevant, so the relation

$$\mathcal{S}_s(\xi) = \mathcal{S}_{\text{JNR}}(\xi) + O\left(\frac{1}{\lambda r^3}\right), \quad |\xi| \lesssim \lambda \quad (3.11)$$

may be enforced by choosing

$$\hat{y}_1 = y_1 + \zeta_1 = y_1 - \frac{\lambda_1^2}{l^2} l_1 = y_1 - \frac{\lambda_1^2}{l^2} (y_1 - z), \quad (3.12a)$$

$$\rho_1^2 = \lambda_1^2 - \frac{2\lambda_1^4}{l^2} - \frac{\lambda_1^2 \lambda_2^2}{r^2}, \quad (3.12b)$$

where we have eliminated higher-order corrections in Eq. (3.12), since they may be subsumed into the correction term of Eq. (3.11). Parallel equations hold for the second instanton at position $\hat{y}_2 = y_2 + \zeta_2$.

Note that Eq. (3.11) is just what would be expected from Eq. (2.7), with Θ_μ of order λ^2/r^3 in the near region. The far region is most easily treated by doing a direct comparison of the gauge potentials of the solution and the superposition. The above calculation of the action density in the near region has given us enough information to guess a gauge in which Θ_μ is small in Eq. (2.7). At the very least, we must make a gauge transformation which removes the singularity at the point z of the JNR solution; furthermore, the instantons of the superposition must have their singularities moved from the points $\hat{y}_1, \hat{y}_2$ to y_1, y_2 where the solution has its singularities.

Now recall that the gauge transformation which removes a gauge singularity of the form $A_\mu = -\bar{\sigma}_{\mu\nu} \partial_\nu \ln 1/x^2$ is

$$g_x \equiv \frac{x_4 - i\vec{\sigma} \cdot \vec{x}}{(x^2)^{1/2}}, \quad (3.13)$$

where $x_\mu = (\vec{x}, x_4)$ and where A_μ , under a gauge transformation g , obeys

$$A_\mu \rightarrow A'_\mu = g^{-1} A_\mu g + i g^{-1} \partial_\mu g. \quad (3.14)$$

The following identities are useful:

$$i g_x^{-1} \partial_\mu g_x = \sigma_{\mu\nu} \partial_\nu \ln x^2, \quad (3.15a)$$

$$g_x^{-1} \bar{\sigma}_{\mu\nu} g_x = \sigma_{\mu\nu} - \frac{2x_\mu x_\nu \sigma_{\alpha\alpha}}{x^2} - \frac{2x_\nu x_\mu \sigma_{\mu\alpha}}{x^2}, \quad (3.15b)$$

$$g_x^{-1} \bar{\sigma}_{\mu\nu} x_\nu g_x = -\sigma_{\mu\nu} x_\nu, \quad (3.15c)$$

where $\sigma_{\mu\nu} = (\sigma_a/2)\eta_{a\mu\nu}$.

The JNR solution, made regular at the point z , is

$$(A_\mu^{\text{JNR}})' = g_{x-z}^{-1} A_\mu^{\text{JNR}} g_{x-z} + i g_{x-z}^{-1} \partial_\mu g_{x-z}. \quad (3.16)$$

A single instanton, with its singularity moved from $\hat{y}_j$ to y_j , is given by

$$(A_\mu^{(j)})' = G_j^{-1} A_\mu^{(j)} G_j + i G_j^{-1} \partial_\mu G_j, \quad (3.17)$$

with

$$G_j \equiv g_{x-\hat{y}_j} g_{x-y_j}^{-1} \quad (3.18)$$

and where $A_\mu^{(j)}$ is of course just

$$\begin{aligned} A_\mu^{(j)} &= A_\mu(\rho_j, \hat{y}_j) \\ &= -\bar{\sigma}_{\mu\nu} \partial_\nu \ln \left[1 + \frac{\rho_j^2}{(x - \hat{y}_j)^2} \right]. \end{aligned} \quad (3.19)$$

Furthermore, note that in zeroth order near y_1 ,

$$h(x) = g_{y_2-z}^{-1} G_2^{-1}(x) G_2(y_1) g_{y_2-z} g_{y_1-z}^{-1} G_1^{-1}(x) g_{y_1-z} g_{y_2-z}^{-1} G_2^{-1}(y_1) g_{y_2-z}. \quad (3.22)$$

Using the fact that

$$G_j(x) = 1 + O(\lambda^2/r^2) \text{ for } |x - y_j| \geq r, \quad (3.23)$$

Eq. (3.20) then becomes

$$(g_{x-z} h)^{-1} A_\mu^{\text{JNR}}(\lambda_i, y_i, z) g_{x-z} h + i (g_{x-z} h)^{-1} \partial_\mu (g_{x-z} h) + \Theta_\mu = \tilde{g}_{y_1-z}^{-1} A_\mu(\rho_1, \hat{y}_1) \tilde{g}_{y_1-z} + \tilde{g}_{y_2-z}^{-1} A_\mu(\rho_2, \hat{y}_2) \tilde{g}_{y_2-z}, \quad (3.24a)$$

where

$$\tilde{g}_{y_1-z} \equiv g_{y_1-z} [g_{y_2-z}^{-1} G_2^{-1}(y_1) g_{y_2-z}] = g_{y_1-z} + O(\lambda^2/r^2), \quad (3.24b)$$

$$\tilde{g}_{y_2-z} \equiv g_{y_2-z} [g_{y_1-z}^{-1} G_1^{-1}(y_2) g_{y_1-z}] = g_{y_2-z} + O(\lambda^2/r^2), \quad (3.24c)$$

and where we have used the same symbol Θ_μ for the correction term since it has the same magnitude as before [Eq. (3.21)]. To derive Eqs. (3.24), we must notice that

$$G_2^{-1}(x) A_\mu(\rho_1, \hat{y}_1) G_2(x) - G_2^{-1}(y_1) A_\mu(\rho_1, y_1) G_2(y_1) = \begin{cases} O(\lambda^2/r^3) & \text{near } \hat{y}_1 \text{ or } \hat{y}_2, \\ O(\lambda^4/r^5) & \text{far from } \hat{y}_1 \text{ and } \hat{y}_2. \end{cases} \quad (3.25)$$

Equations (3.24) together with Eqs. (3.12) and (3.21) agree precisely with the general Eqs. (2.7) and (2.22): An exact solution plus a small correction term is equal to a superposition of two singular gauge instantons with given gauge orientation. Note that Θ_μ in Eq. (3.21) does not satu-

$(A_\mu^{\text{JNR}})'$ looks like a single instanton with group orientation $g = g_{y_1-z}$. We are thus led to try the following relation between the gauge transformed solution and a superposition:

$$\begin{aligned} &g_{x-z}^{-1} A_\mu^{\text{JNR}}(\lambda_i, y_i, z) g_{x-z} + i g_{x-z}^{-1} \partial_\mu g_{x-z} + \Theta_\mu \\ &= \sum_{j=1}^2 g_{y_j-z}^{-1} [G_j^{-1} A_\mu(\rho_j, \hat{y}_j) G_j + i G_j^{-1} \partial_\mu G_j] g_{y_j-z}, \end{aligned} \quad (3.20)$$

where the parameters are related by Eq. (3.12). A straightforward but tedious calculation in which one expands out both sides of Eq. (3.20) and makes heavy use of Eq. (3.15) then shows that this was the right guess:

$$\Theta_\mu \approx \begin{cases} \frac{\lambda^2}{r^3} & \text{near region } (|x - y_i| \leq \lambda \text{ for either } y_i) \\ \frac{\lambda^4}{r^5} & \text{far region } (|x - y_i| \geq r \text{ for both } y_i). \end{cases} \quad (3.21)$$

Equation (3.20) is not precisely in the form of Eq. (2.7), in that the instantons of the superposition have had their singularities moved. It is, however, expressed in a gauge in which the relation between superposed instantons and an exact solution is simplest to guess. To put it in the form of Eq. (2.7) requires a gauge transformation h given by

rate the bound in the far region of Eq. (2.22); therefore it must saturate the bound in the near region. The results of Sec. II guarantee that there is no clever choice of the parameters or gauge of the exact solution which would make Θ_μ smaller in both regions.

Since Eq. (3.24) relates the superposition of two instantons to an exact solution in a special gauge, the reader may wonder how much of the information contained in it has gauge-invariant meaning. Certainly the positions and scales of the two instantons are physically meaningful—they were obtained by a gauge-invariant calculation of the action density. The SU(2) orientations of the two instantons $\tilde{g}_{y_1-z}$ and $\tilde{g}_{y_2-z}$ are of course individually meaningless, since they can be changed by an overall gauge rotation of both sides of Eq. (3.24). On the other hand, it is claimed that the relative SU(2) orientation of the two instantons $\tilde{g}_{y_2-z}\tilde{g}_{y_1-z}^{-1}$ has gauge-invariant significance. This is a non-trivial statement; *a priori*, it would seem possible to make a position dependent gauge transformation on the exact solution which rotates the field near one instanton differently from the field near the other, thus making it correspond to a superposition with different relative orientation. However, such a gauge transformation would have an inhomogeneous term which would induce corrections to Eq. (3.24) which are larger than Θ_μ in the far region (between the two instantons). We can see this by looking at the gauge-invariant action density in the far region. If we superpose two instantons $g_1^{-1}A^{(1)}g_1$ and $g_2^{-1}A^{(2)}g_2$, with relative orientation $g=g_2g_1^{-1}$, then the action density in the far region is

$$S_s = \text{tr}(F_{\mu\nu}^{(1)})^2 + \text{tr}(F_{\mu\nu}^{(2)})^2 + 2\text{tr}(\mathfrak{F}_{\mu\nu}^{(1)}g^{-1}\mathfrak{F}_{\mu\nu}^{(2)}g) + O\left(\frac{\lambda^6}{r^{10}}\right), \quad (3.26)$$

where $\mathfrak{F}_{\mu\nu}^{(i)}$ is defined in Eq. (2.11). On the other hand, the action density of the solution A_μ^g plus the correction Θ_μ in the far region is

$$S = \text{tr}(F_{\mu\nu}^g)^2 + 4\partial_\mu[F_{\mu\nu}^g\Theta_\nu] + \text{higher order in } \Theta_\mu, \quad (3.27)$$

where the term linear in Θ_μ is a total divergence which would give no contribution to the total action. Since $\mathfrak{F}_{\mu\nu}^{(i)}$ falls like λ^2/r^4 for $|x-y_i| \sim r$, a change in g of order 1 in Eq. (3.26) induces a change of order λ^4/r^8 in the action density of the superposition. Examination of Eq. (3.27) then shows that this would require a change in Θ_μ of order λ^2/r^3 —larger by r^2/λ^2 than the old Θ_μ . Thus, by requiring that Θ_μ be as small as possible, we fix the order 1 part of g , which therefore has physical meaning. On the other hand, shifts in g of order λ^2/r^2 can be absorbed into Θ_μ without changing its magnitude; in particular, by Eq. (3.24) there is no physical distinction between $\tilde{g} = \tilde{g}_{y_2-z}\tilde{g}_{y_1-z}^{-1}$ and $g = g_{y_2-z}g_{y_1-z}^{-1}$. This means there must exist a slightly different gauge for Eq. (3.24) in which $\tilde{g}_{y_1-z}$ is replaced by g_{y_1-z} and the magni-

tude of Θ_μ is unchanged.

We can now understand, in the dilute-gas limit, the meaning of the parameter z of the JNR solution. Let us choose, for convenience, our coordinate system so that $\hat{y}_2 = -\hat{y}_1 = (a, \vec{0})$ and $z = (0, \vec{z})$. Then the relative SU(2) orientation of the two instantons in the superposition is given by

$$g \simeq g_{y_2-z}g_{y_1-z}^{-1} = g_{\hat{y}_2-z}g_{\hat{y}_1-z}^{-1} = -\frac{a^2 + \vec{z}^2 + 2ia\vec{z} \cdot \vec{\sigma}}{a^2 + z^2}, \quad (3.28)$$

where we have used Eqs. (3.12) and (3.13) and must remember not to take seriously variations in g of order λ^2/r^2 . As z runs over its possible values on the hyperplane bisecting the line joining $\hat{y}_2$ and $\hat{y}_1$ (to the order which we can determine z from g this is the same as the hyperplane bisecting the line joining y_2 and y_1), g sweeps over all of SU(2). In particular $|\vec{z}| = \infty$ is the 't Hooft solution, $g=1$. Keeping $\hat{y}_i$ and ρ_i fixed by adjusting λ_i, y_i through Eq. (3.12), we produce all solutions representing two instantons of given scale and position and arbitrary orientation. A point raised at the beginning of this section can now be cleared up. If we fix $\rho_i, \hat{y}_i$, and g , and look for the JNR solutions to which this situation corresponds, making no restriction on the location of z , then it is not hard to see—to the accuracy to which z can be determined—that all possible values of z, y_1 , and y_2 lie on a circle. This is just the one-parameter family of equivalences of the JNR solution. (See Appendix A for further discussion of these equivalences.)

Finally, we note that a similar interpretation of parameters can be made for the 't Hooft solution of topological charge k :

$$A_\mu = -\bar{\sigma}_{\mu\nu}\partial_\nu \ln \Pi, \quad (3.29a)$$

$$\Pi = 1 + \sum_{i=1}^k \frac{\lambda_i^2}{(x-y_i)^2}. \quad (3.29b)$$

Making the correspondence is in fact much easier than for the JNR $k=2$ solution, because there is no "renormalization" of the positions and hence no subtleties due to the shifting of gauge singularities. All the instantons are aligned ($g_i=1$); corresponding to Eq. (3.12) is

$$\hat{y}_i = y_i, \quad (3.30a)$$

$$\rho_i^2 = \lambda_i^2 - \lambda_i^2 \sum_{j \neq i} \frac{\lambda_j^2}{(y_j - y_i)^2}. \quad (3.30b)$$

IV. ONE-LOOP QUANTUM CORRECTIONS

The understanding of the classical many-instanton solutions achieved in the previous sections can now be used to gain some insight into the cal-

culation of the first quantum corrections to the classical action. There are two parts to such a calculation: (a) the determinant, in the space of nonzero modes, of the operator which appears in the quadratic part of the expansion of the action about a solution, and (b) the Jacobian which appears in the change of variables from zero modes to collective coordinates. To be specific, in a pure SU(2) Yang-Mills theory, the one-loop contribution to the vacuum-to-vacuum amplitude due to fluctuations about the k -instanton solution is given by

$$Z_k = \int \prod_{n=1}^{8k} d\gamma_n J(\gamma) \exp \left[-\frac{8\pi^2 k}{g_0^2} + W_G(\gamma) + W_\varepsilon(\gamma) \right], \quad (4.1)$$

where g_0 is the coupling constant, γ_n are the $8k$ collective modes (three of which correspond to an overall gauge rotation²), $J(\gamma)$ is the Jacobian described above, and W_G and W_ε are logarithms of the gauge and ghost one-loop determinants, respectively. They have the form¹⁰

$$W_G(\gamma) = -\frac{1}{2} \ln \det' G^{-1}(\gamma), \quad (4.2a)$$

$$W_\varepsilon(\gamma) = \ln \det \Delta^{-1}(\gamma), \quad (4.2b)$$

where $G(\gamma)$ and $\Delta(\gamma)$ are the propagators for the gauge and ghost fields about the k -instanton solution described by collective coordinates γ_n . The prime on the determinant in Eq. (4.2a) means that it is calculated in the space orthogonal to the zero modes.

Let us concentrate first on the one-loop determinant. For the 't Hooft solution of topological charge k , (Eq. 3.29), W_G and W_ε are calculated in Ref. 10:

$$W_G + W_\varepsilon = -\frac{1}{144\pi^2} \int d^4x \frac{(\partial_\mu \partial_\nu \Pi)^2}{\Pi^2} + \text{const} + \Gamma, \quad (4.3)$$

where the constant is independent of the parameters λ_i, y_i and serves to subtract the ultraviolet singularities of the integration which occur at each y_i , and where Γ depends on the coefficients $c_{rs,tu}$ and the functions $\Phi_{rsa}(x)$ which appear in the isospin-one scalar propagator.¹⁵ In the dilute-gas limit, it will turn out that Γ is smaller by at least a factor of λ^2/r^2 than the terms we will be interested in.

An examination of the integral in Eq. (4.3) in the dilute-gas limit ($\lambda_i \sim \lambda$, $|y_i - y_j| \sim r$) shows that it has the form

$$\int d^4x \frac{(\partial_\mu \partial_\nu \Pi)^2}{\Pi^2} = \text{const} + a_1 \ln \lambda^2 + a_2 \frac{\lambda^2}{r^2} + \dots \quad (4.4)$$

The methods used in deriving an expression such as Eq. (4.4) and calculating the coefficients a_1 and a_2 will be presented shortly. The $\ln \lambda^2$ term in Eq. (4.4) contributes to a renormalization of the coupling constant to mass scale $1/\lambda$ (see Ref. 2); the λ^2/r^2 term seems to represent a Coulomb interaction among instantons in 4 dimensions.¹⁶ If present, such a term might result in a breakdown of the dilute-gas approximation (a phase transition or at least strong density correlations).

To investigate the Coulomb term further, and to check for possible orientation dependence of the interaction, we turn to the complete solution for two instantons, namely to the JNR $k=2$ solution discussed in Sec. III. Our first problem is to generalize the results of Ref. 10 from the 't Hooft solution, Eq. (3.29), to the JNR solution, Eqs. (3.1) and (3.2). Appendix B shows that if we are not interested in terms of order λ^4/r^4 , we can neglect Γ in Eq. (4.3) and replace Π by Π^{JNR} :

$$W_G + W_\varepsilon = -\frac{1}{144\pi^2} \int d^4x \frac{(\partial_\mu \partial_\nu \Pi^{\text{JNR}})^2}{(\Pi^{\text{JNR}})^2} + \text{const} + O(\lambda^4/r^4), \quad (4.5)$$

where the constant now must subtract an additional infrared (large x) divergence.

Extraction of the terms of interest from Eq. (4.5) is tedious but not difficult. First, let us write

$$(\partial_\mu \partial_\nu \Pi^{\text{JNR}})^2 = \sum_{i,j=0}^2 M_{ij}, \quad (4.6a)$$

$$M_{ij} = \frac{16\lambda_i^2 \lambda_j^2}{(x-y_i)^4 (x-y_j)^4} \times \left[\frac{4|(x-y_i) \cdot (x-y_j)|^2}{(x-y_i)^2 (x-y_j)^2} - 1 \right], \quad (4.6b)$$

$$P_{ij} \equiv \int d^4x \frac{M_{ij}}{(\Pi^{\text{JNR}})^2} \quad (4.6c)$$

with the notation $\lambda_0 = l$, $y_0 = z$. All the P_{ij} diverge for large x ; the diagonal P_{ii} also diverge for small $|x - y_i|$. The sum of these divergences is parameter independent; however, it is easier to work with each term separately. We now introduce a much simpler set of quantities K_{ij} which have the same divergences as P_{ij} :

$$K_{00} = 48 \int_0^\lambda \frac{d^4x}{x^4} + \frac{48\lambda_0^4}{\left(\sum_j \lambda_j^2\right)^2} \int_\lambda^\infty \frac{d^4x}{x^4}, \quad (4.7a)$$

$$K_{ii} = 48 \int_0^{\lambda_i} \frac{d^4x}{x^4} + \frac{48\lambda_i^4}{\left(\sum_j \lambda_j^2\right)^2} \int_{\lambda_i}^\infty \frac{d^4x}{x^4}, \quad i \neq 0 \quad (4.7b)$$

$$K_{i0} = K_{0i}$$

$$= \frac{16\lambda_0^2 \lambda_i^2}{\left(\sum_j \lambda_j^2\right)^2} \int_{\lambda_i}^{\infty} \frac{d^4 x}{x^4} \left\{ \frac{4[x \cdot (x + l_i)]^2}{x^2(x + l_i)^2} - 1 \right\}, \quad i \neq 0 \quad (4.7c)$$

$$K_{12} = K_{21}$$

$$= \frac{16\lambda_1^2 \lambda_2^2}{\left(\sum_j \lambda_j^2\right)^2} \int_{\lambda}^{\infty} \frac{d^4 x}{x^4} \left[\frac{4|x \cdot (x + r)|^2}{x^2(x + r)^2} - 1 \right], \quad (4.7d)$$

where λ , r , and l_i are given by Eqs. (3.3) and (3.4), and all integrations are done over spheres of the indicated radii. The finite limits in Eqs. (4.7) have been chosen for later convenience; we introduce λ (rather than use λ_1 or λ_2) to maintain the symmetry between instantons 1 and 2.

We now define a set of finite quantities, P'_{ij} :

$$P'_{ij} \equiv P_{ij} - K_{ij}. \quad (4.8)$$

Shifts in the integration variable of P_{ij} (legitimate because all divergences are logarithmic) will be necessary to make the finiteness of P'_{ij} manifest.

The evaluation of P'_{ij} is fairly straightforward. The following identities for angular integrations are useful:

$$\int d\Omega_x = 2\pi^2, \quad (4.9a)$$

$$\int d\Omega_x (x \cdot w) = 0, \quad (4.9b)$$

$$\frac{1}{2\pi^2} \int d\Omega_x (x \cdot w)^2 = \frac{x^2 w^2}{4}, \quad (4.9c)$$

$$\frac{1}{2\pi^2} \int d\Omega_x \frac{1}{(x - w)^2} = \begin{cases} \frac{1}{x^2}, & |x| > |w| \\ \frac{1}{w^2}, & |x| < |w| \end{cases} \quad (4.9d)$$

where Ω_x is the angular orientation of the vector x_μ , and w is an arbitrary 4-vector. Equation (4.9d) is just the potential of a spherical charge distribution of radius $|x|$ in four dimensions.

We now calculate P'_{00} up to $O(\lambda^4/r^4)$ as an example. Using Eqs. (4.6), (4.7), and (3.2) we have, after a shift in the integration variable of P_{00} , $x \rightarrow x + z$:

$$P'_{00} = P'_{00}^{(a)} + P'_{00}^{(b)}, \quad (4.10a)$$

$$P'_{00}^{(a)} = 48 \int_0^\lambda \frac{d^4 x}{x^4} \left\{ \frac{l^4}{[l^2 + \lambda_1^2 x^2 / (x - l_1)^2 + \lambda_2^2 x^2 / (x - l_2)^2]^2} - 1 \right\}, \quad (4.10b)$$

$$P'_{00}^{(b)} = 48 \int_\lambda^\infty \frac{d^4 x}{x^4} \left\{ \frac{l^4}{[l^2 + \lambda_1^2 x^2 / (x - l_1)^2 + \lambda_2^2 x^2 / (x - l_2)^2]^2} - \frac{l^4}{(l^2 + \lambda_1^2 + \lambda_2^2)^2} \right\}. \quad (4.10c)$$

Finding common denominators and neglecting terms with explicit λ_i^4 gives

$$P'_{00}^{(a)} = -96l^2 \int_0^\lambda \frac{d^4 x}{x^2} \frac{\sum_{i=1}^2 \frac{\lambda_i^2}{(x - l_i)^2}}{\left[l^2 + \sum_{i=1}^2 \frac{\lambda_i^2 x^2}{(x - l_i)^2} \right]^2} + O\left(\frac{\lambda^4}{l^4}\right), \quad (4.11a)$$

$$P'_{00}^{(b)} = 96l^2 \int_\lambda^\infty \frac{d^4 x}{x^4} \frac{\sum_{i=1}^2 \lambda_i^2 \left(1 - \frac{x^2}{(x - l_i)^2} \right)}{\left[l^2 + \sum_{i=1}^2 \frac{\lambda_i^2 x^2}{(x - l_i)^2} \right]^2} + O\left(\frac{\lambda^4}{l^4}\right), \quad (4.11b)$$

where we have used Eq. (3.3). Since the numerators already have explicit factors of λ_i^2 (the regions $|x - l_i| \sim \lambda$ are unimportant), we may drop the terms $\lambda_i^2 x^2 / (x - l_i)^2$ compared to l^2 in the denominators. $P'_{00}^{(a)}$ is then clearly of order λ^4/l^4 , while $P'_{00}^{(b)}$ becomes

$$P'_{00}^{(b)} = \frac{96}{l^2} \sum_{i=1}^2 \lambda_i^2 \int_\lambda^\infty \frac{d^4 x}{x^4} \left[1 - \frac{x^2}{(x - l_i)^2} \right] + O\left(\frac{\lambda^4}{l^4}\right). \quad (4.12)$$

Using Eqs. (4.9a) and (4.9d), we have

$$P'_{00} = 96\pi^2 (\ln l^2 / \lambda^2 - 1) \sum_{i=1}^2 \frac{\lambda_i^2}{l^2} + O\left(\frac{\lambda^4}{r^4}\right), \quad (4.13)$$

where we have used $\lambda^4/l^4 \lesssim \lambda^4/r^4$. Similar calculations give

$$P'_{11} = 48\pi^2 \left(\frac{\lambda_1^2}{l^2} - \frac{\lambda_2^2}{r^2} \right) + O\left(\frac{\lambda^4}{r^4}\right), \quad (4.14a)$$

$$P'_{22} = 48\pi^2 \left(\frac{\lambda_2^2}{l^2} - \frac{\lambda_1^2}{r^2} \right) + O\left(\frac{\lambda^4}{r^4}\right), \quad (4.14b)$$

$$P'_{01} \sim P'_{02} \sim P'_{12} \sim \lambda^4/r^4. \quad (4.14c)$$

We still need to evaluate the K_{ij} . For $i=j$ the integrals are trivial; for $i \neq j$ the integrals are made trivial by using Eq. (4.9d) and the identity

$$x \cdot (x + \omega) = [x^2 - \omega^2 + (x - \omega)^2]/2. \quad (4.15)$$

The result is

$$\sum_{i,j} K_{ij} = 48\pi^2 \left[\ln \lambda_1^2 + \ln \lambda_2^2 + \left(2 \ln \frac{\lambda^2}{l^2} - 1 \right) \left(\frac{\lambda_1^2}{l^2} + \frac{\lambda_2^2}{l^2} \right) \right] + O\left(\frac{\lambda^4}{r^4}\right) + \text{const}, \quad (4.16)$$

where the const is infinite but parameter independent, as promised. Equations (4.5), (4.6), (4.8), (4.13), (4.14), and (4.16) now give

$$W_G + W_\epsilon = -\frac{1}{3} \left[\sum_{i=1}^2 \ln \lambda_i^2 - \sum_{i=1}^2 \lambda_i^2 \left(\frac{1}{r^2} + \frac{1}{l^2} \right) \right] + \text{const} + O\left(\frac{\lambda^4}{r^4}\right). \quad (4.17)$$

For two infinitely separated instantons ($r^2, l^2 \rightarrow \infty$) this agrees with the calculation of Ref. 2.

Equation (4.7) would seem to represent a Coulomb-type, orientation-dependent [through $l^2 = (y_i - z)^2$] potential between instantons. However, recall that the scales λ_i which appear in Eq. (4.17) are not the scales of the superposed instantons to which the JNR $k=2$ solution corresponds. From Eq. (3.12), the λ_i are expressed in terms of the "true" scales ρ_i by

$$\lambda_1^2 = \rho_1^2 + \frac{\rho_1^2 \rho_2^2}{r^2} + \frac{2\rho_1^4}{l^2} + O\left(\frac{\lambda^4}{r^4}\right), \quad (4.18a)$$

$$\lambda_2^2 = \rho_2^2 + \frac{\rho_1^2 \rho_2^2}{r^2} + \frac{2\rho_2^4}{l^2} + O\left(\frac{\lambda^4}{r^4}\right). \quad (4.18b)$$

When Eq. (4.18) is used in Eq. (4.17), we have

$$W_G + W_\epsilon = -\frac{1}{3} \sum_{i=1}^2 \ln \rho_i^2 + \text{const} + O\left(\frac{\lambda^4}{r^4}\right). \quad (4.19)$$

The Coulomb interaction has entirely disappeared. We are saying that the one-loop action respects, at this order, the invariance of the classical action under changes in instanton separation. Several remarks are appropriate here:

(1) With hindsight, Eq. (4.19) is perhaps not very surprising. One is merely saying that in the widely separated limit, the "widely separated parameters," i.e., the parameters of the superposition, are the appropriate ones to use. Our result may be taken as an upper bound for the error (at one loop) one makes in approximating the many-instanton sector as the iteration of the one-instanton sector, i.e., in performing the calculation for an infinitely dilute gas. If one were clever enough to implement in detail the prescription of Ref. 4 for computing quantum fluctuations

about an approximate solution (the superposition of instantons), one might be able to arrive at Eq. (4.19) more directly.

(2) Equation (4.19), by itself, is not the whole story, because we have not yet investigated the collective-coordinate Jacobian $J(\gamma)$ which also appears in Eq. (4.1). We will show presently, however, that when the exact solution is parametrized in terms of the superposition, terms of order λ^2/r^2 also disappear from $J(\gamma)$.

(3) Since an expression such as Eq. (4.3) holds in addition for massless scalars or spin- $\frac{1}{2}$ fermions of isospin $\frac{1}{2}$ or 1 (the coefficient in front of the integral varies, and Γ is absent for isospin $\frac{1}{2}$), our result is not confined to a pure Yang-Mills theory. Of course, Eq. (4.3) gives the determinant over nonzero modes only; when massless fermions are present their zero modes will cause the entire amplitude to vanish unless we calculate a Green's function with enough external lines to fill these states.

(4) A similar calculation can be done for arbitrary instanton number if the instantons are aligned ('t Hooft solution). The result is

$$W_G + W_\epsilon = -\frac{1}{3} \left[\sum_{i=1}^k \ln \lambda_i^2 - \sum_i \sum_{j \neq i} \frac{\lambda_i^2}{(y_j - y_i)^2} \right] + \text{const} + O\left(\frac{\lambda^4}{r^4}\right). \quad (4.20)$$

The redefinition of parameters is given by Eq. (3.30); corresponding to Eq. (4.18) is

$$\lambda_i^2 = \rho_i^2 + \rho_i^2 \sum_{j \neq i} \frac{\rho_j^2}{(y_j - y_i)^2} + O\left(\frac{\lambda^4}{r^4}\right), \quad (4.21)$$

which again eliminates the λ^2/r^2 interaction when

inserted into Eq. (4.20).

(5) Higher-order corrections to Eq. (4.17) are computable in principle but become increasingly tedious. We have no reason to suspect that higher-order interactions can be made to vanish. (See the discussion at the end of this section.)

We turn, finally, to the investigation of the Jacobian $J(\gamma)$ in Eq. (4.1). Consider the zero modes $\psi_\mu^E(n)$ ($n=1, \dots, 8k$) about an exact, topological charge- k solution A_μ^E which depends on $8k$ parameters γ_n . If we work in background gauge $D_\mu^E A_\mu = 0$ then the zero modes are given by

$$\psi_\mu^E(\gamma_n) = \frac{\partial A_\mu^E}{\partial \gamma_n} + D_\mu^E \Lambda_n, \quad (4.22)$$

where Λ_n is the gauge transformation necessary to enforce $D_\mu^E \psi_\mu^E = 0$. The Jacobian then has the form¹⁷

$$J(\gamma) = \left[\det \int d^4x \text{tr} \psi_\mu^E(\gamma_n) \psi_\mu^E(\gamma_m) \right]^{1/2}, \quad (4.23)$$

where the determinant is taken over the indices n and m , and the trace is taken over SU(2) indices. We have suppressed a parameter-independent factor which depends on the coupling constant and the regulator mass.² We wish to show that when the parameters of the corresponding superposition are taken for γ_n , $J(\gamma)$ has the form

$$J(\gamma) = c \prod_{i=1}^k \rho_i^3 \left[1 + O\left(\frac{\lambda^3}{r^3}\right) \right], \quad (4.24)$$

where ρ_i is the scale of the i th instanton in the superposition and c is a parameter-independent constant. The absence of the usual factor of ρ_i^{-5} in Eq. (4.24) should not worry the reader; that factor arises after conventional redefinition of the one-loop determinant.

Before proceeding, we must study the background gauge zero modes of an isolated, singular-gauge instanton. An instanton with position $\hat{y}$, scale ρ , and SU(2) orientation g has a potential given by

$$A_\mu = -\bar{\sigma}_{\mu\nu}^E \partial_\nu \ln(1 + \rho^2/w^2), \quad (4.25)$$

where $\bar{\sigma}_{\mu\nu}^E \equiv g^{-1} \bar{\sigma}_{\mu\nu} g$, and $w \equiv x - \hat{y}$. By gauge transforming the zero modes in Ref. 2 to a singular gauge, we have

$$\psi_\mu(\hat{y}_\nu) = -\frac{4\rho^2}{(\rho^2 + w^2)^2} \left(\bar{\sigma}_{\mu\nu}^E - \frac{2w_\mu w_\nu}{w^2} \bar{\sigma}_{\alpha\nu}^E - \frac{2w_\nu w_\alpha}{w^2} \bar{\sigma}_{\mu\alpha}^E \right), \quad (4.26a)$$

$$\psi_\mu(\rho) = \frac{4\rho \bar{\sigma}_{\mu\nu}^E w_\nu}{(\rho^2 + w^2)^2}, \quad (4.26b)$$

$$\psi_\mu(a) = \frac{2\rho^2}{(\rho^2 + w^2)^2} (\sigma_a^E w_\mu + \epsilon_{abc} \sigma_b^E \bar{\eta}_{c\mu\nu} w_\nu), \quad (4.26c)$$

where $\sigma_a^E (\equiv g^{-1} \sigma_a g)$ are the rotated Pauli matrices, and where the argument of ψ_μ indicates the mode we are considering [$\psi_\mu(a)$ are the three gauge modes]. The modes are orthogonal and obey $D_\mu \psi_\mu = 0$. If we describe the orientation $\hat{g}$ of the instanton in the neighborhood of g by the infinitesimal parameters t_a ,

$$\hat{g} = g(1 - it_a \sigma_a^E), \quad (4.27)$$

then we may put Eq. (4.26) in the form of Eq. (4.22). That is, with the following gauge transformations

$$\Lambda_{\hat{y}_\nu} = \frac{2\rho^2 \bar{\sigma}_{\nu\alpha}^E w_\alpha}{w^2(w^2 + \rho^2)}, \quad (4.28a)$$

$$\Lambda_\rho = 0, \quad (4.28b)$$

$$\Lambda_{t_a} = -\frac{\sigma_a^E \rho^2}{\rho^2 + w^2}, \quad (4.29a)$$

the zero modes take the form

$$\psi_\mu(\gamma_m) = \frac{\partial A_\mu}{\partial \gamma_m} + D_\mu \Lambda_m \quad (4.29b)$$

with $\gamma_m = \hat{y}_\mu, \rho$, or t_a . Finally, note that the Jacobian for this single instanton can be calculated from Eq. (4.23). The result is

$$J_1 = c_1 \rho^3, \quad (4.30)$$

where c_1 is a constant independent of the parameters γ_m . The factor of ρ^3 comes from the three gauge modes, Eq. (4.26c), and is obvious by dimensional analysis.

We can now study the Jacobian of the exact solution of topological charge k for large separation. The basic idea of the discussion is very simple: Since the solution is close to a superposition of individual instantons, the exact zero modes must be close to the zero modes of individual instantons. The only subtlety will, as usual, be a gauge question. We need the exact zero modes in the background gauge of the solution¹⁸; we are given individual zero modes in the background gauge of an individual instanton.

Differentiating Eq. (2.7) with respect to the m th parameter of the i th superposed instanton γ_m^i gives

$$\frac{\partial A_\mu^E}{\partial \gamma_m^i} = \frac{\partial A_\mu^{(i)}}{\partial \gamma_m^i} - \frac{\partial \Theta_\mu}{\partial \gamma_m^i}, \quad (4.31)$$

where, in accordance with our usual philosophy, we use the parameters of the superposition (γ_m^i) as the parameters of the exact solution. The left-hand side of Eq. (4.31) is an exact zero mode, but it is not in the right gauge. Let Λ_{mi}^E be the gauge transformation which puts it in the background gauge of the solution

$$\begin{aligned}\psi_\mu^E(\gamma_m^i) &= \frac{\partial A_\mu^E}{\partial \gamma_m^i} + D_\mu^E \Lambda_{mi}^E, \\ D_\mu^E \psi_\mu^E &= 0,\end{aligned}\quad (4.32)$$

and let Λ_{mi} be one of the gauge transformations, Eq. (4.28), that put $\partial A_\mu^{(i)}/\partial \gamma_m^i$ in the background gauge of $A_\mu^{(i)}$. Then we have

$$\psi_\mu^E(\gamma_m^i) = \psi_\mu(\gamma_m^i) + \chi_\mu(\gamma_m^i), \quad (4.33a)$$

where

$$\chi_\mu(\gamma_m^i) \equiv D_\mu^E \Lambda_{mi}^E - D_\mu^{(i)} \Lambda_{mi} - \frac{\partial \Theta}{\partial \gamma_m^i}. \quad (4.33b)$$

We wish to estimate the size of $\chi_\mu(\gamma_m^i)$. First write

$$\chi_\mu = D_\mu^E(\Delta\Lambda) - i[\Delta A_\mu, \Lambda] - \frac{\partial \Theta}{\partial \gamma}, \quad (4.34)$$

where $\Delta\Lambda \equiv \Lambda^E - \Lambda$, $\Delta A_\mu = A_\mu^E - A_\mu^{(i)}$, and the indices m, i have been dropped for notational simplicity. Equations (4.33) and (4.34) allow us to write, after some manipulation,

$$\begin{aligned}D_\mu^E D_\mu^E(\Delta\Lambda) &= D_\mu^E \frac{\partial \Theta}{\partial \gamma} + i[\Delta A_\mu, \psi_\mu] + [A_\mu^E, [\Delta A_\mu, \Lambda]] \\ &\quad - i[\partial_\mu \Theta, \Lambda] + i[\Delta A_\mu, \partial_\mu \Lambda],\end{aligned}\quad (4.35)$$

where we have used $D_\mu^E \psi_\mu^E = 0$ and $D_\mu \psi_\mu = 0$. Using the known magnitudes of A_μ^E , ΔA_μ , Θ_μ (2.22), ψ_μ (4.26), and Λ (4.28), the magnitude of the right-hand side of Eq. (4.35) can be estimated in the three relevant regions (a) near instanton (i) (b) near some other instanton, (c) far from all instantons. The result is

$$D_\mu^E D_\mu^E(\Delta\Lambda) \lesssim \begin{cases} 1/r^3 & \text{in (a),} \\ 1/r^2 & \text{in (b),} \\ 1/r^5 & \text{in (c),} \end{cases} \quad (4.36)$$

where we have taken the worst case (the gauge modes) for the estimate, and have temporarily set $\lambda = 1$ to avoid the notational complexity caused by the fact that different modes have different dimensions. The operator $D_\mu^E D_\mu^E$ itself has zero modes [the three gauge modes for the exact solution corresponding to Eq. (4.26c) for an individual instanton], so the magnitude of $\Delta\Lambda$ cannot be read off immediately from Eq. (4.36). The zero-mode ambiguities in $\Delta\Lambda$ are treated exactly like the ambiguities of Θ_μ in Sec. II: we absorb all zero modes in $\Delta\Lambda$ into Λ^E just as we absorbed all zero modes in Θ_μ into A_μ^E . Now $\Delta\Lambda$ can be estimated from Eq. (4.36), which then gives an estimate for $D_\mu^E(\Delta\Lambda)$:

$$D_\mu^E(\Delta\Lambda) \lesssim \begin{cases} 1/r^3 & \text{in (a),} \\ 1/r^2 & \text{in (b),} \\ 1/r^4 & \text{in (c),} \end{cases} \quad (4.37)$$

where we have used $D_\mu^E \sim 1$ in (a) and (b) and $D_\mu^E \sim 1/r$ in (c). Equation (4.37) combined with Eq. (4.34) gives the needed result

$$\begin{aligned}\psi_\mu^E(\gamma_m^i) - \psi_\mu(\gamma_m^i) &= \chi_\mu(\gamma_m^i) \\ &\lesssim \begin{cases} 1/r^3 & \text{near instanton (i),} \\ 1/r^2 & \text{near other instantons,} \\ 1/r^4 & \text{far from all instantons.} \end{cases}\end{aligned}\quad (4.38)$$

From Eq. (4.38) it is a simple matter to estimate the size of corrections to the Jacobian (4.23). Using the known behavior of $\psi_\mu(\gamma_m^i)$ given by Eq. (4.26) and adding the contributions from the three regions of Eq. (4.38) to the integration results in

$$\text{tr} \int d^4x \psi_\mu^E(\gamma_m^i) \psi_\mu^E(\gamma_{m'}^{i'}) \sim \begin{cases} \text{const} + 1/r^3, & i=i', m=m' \\ 1/r^3, & i=i', m \neq m' \\ 1/r^2, & i \neq i' \end{cases} \quad (4.39)$$

where we have temporarily set all scales = 1 and have used the orthogonality of $\psi_\mu(\gamma_m^i)$ and $\psi_\mu(\gamma_{m'}^{i'})$ for $m \neq m'$. The $1/r^2$ behavior for $i \neq i'$ comes from the behavior of χ_μ in the second region of Eq. (4.38). Since the off-diagonal matrix elements must contribute in pairs to the determinant, the major contribution comes simply from the diagonal ($i=i', m=m'$) terms. The factors of the scales may then be reinserted by dimensional analysis, and we arrive at Eq. (4.24),

$$J(\gamma) = C \prod_{i=1}^n \rho_i^3 \left[1 + O\left(\frac{\lambda^3}{r^3}\right) \right]. \quad (4.24)$$

Presumably, corrections come only in integral powers of λ^2/r^2 , and the λ^3/r^3 turns into λ^4/r^4 by some symmetric integration, but we have not shown this. Note that Eq. (4.24) is true for arbitrary instanton number, since the derivation relied only on the general results of Sec. II.

We have delayed until now the discussion of a subtle point about the physical meaning of our results. We have shown that with a natural definition of the parameters of the exact solution, the λ^2/r^2 interaction between instantons vanishes. The question is whether one can make all higher orders in the interaction vanish by a clever choice of parameters. Phrased differently, we wish to know whether there is any physical, parameter-independent meaning to the instanton interaction. Unfortunately, this question, like so many others

about unbroken Yang-Mills theories, cannot be rigorously answered until we understand how to solve the infrared problems of the theory. The one-loop calculation presented here cannot be taken completely seriously, for when we try to evaluate the integral over collective coordinates, it will diverge for large scale size. As long as the integral has significant contributions from large instantons, the interactions of small instantons can be made to vanish by an appropriate redefinition of scale which shifts these effects to larger sizes. On the other hand, when we correctly calculate a physical quantity then the integral will be cut off at a certain scale. (The scale will presumably come from higher-loop corrections, or from an intrinsic scale in the problem—e.g., the typical separation of the points of a Green's function.) If that scale is small enough that the dilute-gas approximation makes sense,¹⁹ then there will be a physical meaning to the interactions between instantons. It then seems extremely plausible that the cutoff, which is a physical quantity, will respect the parameters ρ_i which determine the scales over which the action density falls near each instanton. The ρ_i are unambiguous in order λ^2/r^2 so the statement "there is no λ^2/r^2 interaction" is a very plausible conclusion from our calculation. In higher order the ρ_i are ambiguous (the solution no longer looks like a superposition) and there is no physical reason to suppose that a particular choice which makes the interactions appear to vanish would be respected by the cutoff. We must expect, therefore, that λ^4/r^4 interactions do exist.²⁰

V. CONCLUSION AND DISCUSSION

Our main results may be summarized as follows:

(1) By very general arguments it is possible to see that a superposition of widely separated singular gauge instantons with arbitrary gauge orientation differs from an exact solution by a small term Θ_μ , which has bounds

$$\Theta_\mu \lesssim \begin{cases} \lambda^2/r^3 \text{ near region } (|x - y_i| \leq \lambda) \\ \text{for some } y_i) \\ \lambda^3/r^4 \text{ far region } (|x - y_i| \geq r \\ \text{for all } y_i). \end{cases} \quad (2.22)$$

Examination of specific solutions shows that the bound in the far region is not saturated and Θ_μ is actually of order λ^4/r^5 there.

(2) One can understand precisely the meaning of the parameters of the JNR $k=2$ solution in the separated limit by relating it to a superposition

of two instantons. If Π_{JNR} is given by

$$\Pi_{\text{JNR}} = \frac{l^2}{(x-z)^2} + \frac{\lambda_1^2}{(x-y_1)^2} + \frac{\lambda_2^2}{(x-y_2)^2}, \quad (3.2)$$

and the gauge and scale freedom of the solution is used to require

$$\begin{aligned} l^2 &= l_1^2 = l_2^2, \\ l_i &\equiv y_i - z, \end{aligned} \quad (3.3)$$

then the solution in the separated limit, $\lambda_1^2 \sim \lambda_2^2 \ll r^2 = (y_1 - y_2)^2$, is most closely approximated by a superposition of two instantons with positions $\hat{y}_i$, scales ρ_i , and relative SU(2) orientation g , with

$$\begin{aligned} \hat{y}_i &= y_i - \frac{\lambda_i^2}{l^2} (y_i - z), \\ \rho_i^2 &= \lambda_i^2 - \frac{2\lambda_i^4}{\rho^2} - \frac{\lambda_i^2 \lambda_j^2}{r^2} \quad (j \neq i), \end{aligned} \quad (3.12)$$

$$g = \frac{1}{l^2} [l_1 \cdot l_2 - i(l_1^0 \vec{l}_2 - l_2^0 \vec{l}_1 + \vec{l}_1 \times \vec{l}_2) \cdot \vec{\sigma}]. \quad (3.28)$$

Equation (3.28) has been generalized for an arbitrary coordinate system.

(3) The first quantum corrections to an exact solution are elucidated by expressing the exact solution in terms of the parameters of the corresponding superposition. In particular, calculations for the general $k=2$ solution and the aligned ('t Hooft) solution for arbitrary k show that these instanton interactions induced by one-loop quantum corrections are smaller than λ^2/r^2 . This is important not only because it provides a bound for the size of corrections to the limit of infinite separation, but also because it makes more likely the absence of strong long-range correlations in a dilute gas of instantons.

As emphasized previously, the methods of this paper cannot treat the more important physical situation—a superposition of instantons and anti-instantons—because there is no exact solution which looks like such a configuration. The method suggested in Ref. 4 to calculate quantum corrections to approximate solutions seems difficult to implement at present; however, when such a scheme is developed it may be checked against our calculation by applying it to a superposition of instantons alone.

Quantum interactions (like the classical interactions) between instantons and anti-instantons may be stronger than between instantons alone, even without the presence of light quarks. Light quarks, of course, produce a logarithmic interaction between instantons and anti-instantons which would overwhelm all inverse-power interactions. However, it has been suggested^{4,21} that

above a certain critical scale size the quarks get a mass by chiral-symmetry breaking and the instanton-anti-instanton pairs are liberated from their logarithmic interaction. At such scales, effects of the type treated here would be important.

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APPENDIX A

We exhibit here two physically equivalent configurations of the three JNR points which differ from one another by a finite translation of the points around the circle joining them. In fact, both these configurations have the property that z is on the hyperplane bisecting the line between y_1 and y_2 , so this example will allow us to elucidate the meaning of the discrete symmetry which remains even after we demand that z be on the hyperplane.

For simplicity we consider the case where the points lie on a straight line (a degenerate circle) and where the two instantons have the same scale. The claim is that the two choices

$$\Pi = \frac{y^2}{x^2} + \frac{\lambda^2}{(x-y)^2} + \frac{\lambda^2}{(x+y)^2} \quad (z=0, l^2=y^2) \quad (\text{A1})$$

and

$$\Pi' = 1 + \frac{\lambda'^2}{(x-y')^2} + \frac{\lambda'^2}{(x+y')^2} \quad (z=\infty, l^2/z^2=1) \quad (\text{A2})$$

result in physically equivalent configurations provided

$$\lambda^2 = \left(1 + \frac{2\lambda'^2}{y'^2}\right) \lambda'^2 \quad (\text{A3})$$

and

$$y_\mu = \left(1 + \frac{2\lambda'^2}{y'^2}\right)^{1/2} y'_\mu. \quad (\text{A4})$$

Note that the second choice (A2) is just the 't Hooft case.

We shall not prove the equivalence of (A1) and (A2) but rather make it extremely plausible by showing that they have identical action densities. Equations (3.5) and (3.6) give for the two action

densities

$$S = \square \square \ln \Pi = \square \square \ln [\Pi x^2 (x-y)^2 (x+y)^2] \quad (\text{A5})$$

and

$$S' = \square \square \ln \Pi' = \square \square \ln [\Pi' (x-y')^2 (x+y')^2]. \quad (\text{A6})$$

Simple algebra now shows that the choices (A3) and (A4) result in

$$\Pi x^2 (x-y)^2 (x+y)^2 = c^2 \Pi' (x-y')^2 (x+y')^2, \quad (\text{A7})$$

where $c^2 = y^4/y'^2$. Thus the action densities are everywhere equal.

Since these two configurations are equivalent, we may check the results of Sec. III by seeing if they correspond to the same superposition in the widely separated limit. Equations (3.12) and (3.28) give the parameters of the superposition in each case:

$$\begin{aligned} \hat{y}_\mu &= \left(1 - \frac{\lambda^2}{y^2}\right) y_\mu, \quad \hat{y}'_\mu = y'_\mu \\ \rho^2 &= \lambda^2 \left(1 - \frac{\lambda^2}{y^2}\right), \quad \rho'^2 = \lambda'^2 \left(1 - \frac{\lambda'^2}{y'^2}\right), \\ g &= 1, \quad g' = -1. \end{aligned} \quad (\text{A8})$$

To the order to which Eq. (3.12) can be taken seriously, we have $\hat{y}_\mu = \hat{y}'_\mu$, $\rho^2 = \rho'^2$. The fact that $g' = -g$ may be disconcerting until we realize that A_μ transforms according to the adjoint representation of SU(2) [i.e., like an O(3) vector], so that g and $-g$ define the same relative orientation of the instantons.

This phenomenon is not confined to the case where y_1 , y_2 , and z lie along a line. By a conformal transformation, or an inversion, of both configurations, we may generate another pair of equivalent configurations all of whose points lie along a circle. It is not very difficult to see that when we make an inversion which leaves z on the hyperplane between y_1 and y_2 , z' will be on the hyperplane between y'_1 and y'_2 . Furthermore, the resulting configurations correspond to the same superposition, with $g' = -g$. Thus, the discrete symmetry that remains in the JNR solution (after the third point is required to be on the hyperplane bisecting the line between the other two) is seen to be a simple reflection of the fact that instanton orientation is specified by an element of O(3) and that SU(2) forms a double covering of O(3).

APPENDIX B

In this appendix we wish to generalize the results of Ref. 10 (and, of necessity, Ref. 15) from the 't Hooft to the JNR solution. Most of the generalization is completely trivial—we must replace

Π everywhere it appears explicitly by Π^{JNR} . The only other change comes in the calculation of the coefficients $c_{rs,tu}$ that appear in the isospin-one propagator and in Γ of Eq. (4.4).

For the 't Hooft case, recall that the isospin-one scalar propagator is written¹⁵ as

$$\Delta_{ab}(x, y) = \frac{W_{ab}(x, y)}{4\pi^2(x-y)^2} + \frac{C_{ab}(x, y)}{4\pi^2\Pi(x)\Pi(y)}, \quad (\text{B1})$$

where the coefficients $c_{rs,tu}$ enter into the function $C_{ab}(x, y)$ through

$$C_{ab}(x, y) = - \sum_{r,s,t,u} \phi_{rsa}^\dagger(x) c_{rs,tu} \phi_{tub}^\dagger(y), \quad (\text{B2})$$

with

$$\begin{aligned} \phi_{rsa}^\dagger(x) &\equiv \frac{\lambda_r \lambda_s}{(x-y_r)^2(x-y_s)^2} \\ &\times \bar{\eta}_{\mu\nu}(x-y_r)_\mu (x-y_s)_\nu. \end{aligned} \quad (\text{B3})$$

The coefficients $c_{rs,tu}$ obey a complicated matrix equation which has the solution

$$\begin{aligned} c_{rs,tu} &= \frac{\delta_{ru}\delta_{sv} - \delta_{rv}\delta_{su}}{(y_r - y_s)^2} \\ &- \frac{1}{(y_r - y_s)^2} (\lambda_r \lambda_u f_{sv} - \lambda_r \lambda_v f_{su} - \lambda_s \lambda_u f_{rv} \\ &\quad + \lambda_s \lambda_v f_{ru}) \frac{1}{(y_u - y_v)^2}, \end{aligned} \quad (\text{B4})$$

where f_{st} is the inverse of the matrix g_{st} defined by

$$\begin{aligned} g_{st} &= \left[1 + \sum_{r \neq s} \frac{\lambda_r^2}{(y_r - y_s)^2} \right] \delta_{st} \\ &- \frac{\lambda_s \lambda_t}{(y_s - y_t)^2} (1 - \delta_{st}). \end{aligned} \quad (\text{B5})$$

In the JNR case, Eqs. (B1)–(B3) remain unchanged (of course Π is Π^{JNR} , and we should let $\lambda_0 = l$, $y_0 = z$). However, the matrix g_{st} is replaced by h_{st} , with

$$\begin{aligned} h_{st} &= \left[\sum_{r \neq s} \frac{\lambda_r}{(y_r - y_s)^2} \right] \delta_{st} \\ &- \frac{\lambda_s \lambda_t}{(y_s - y_t)^2} (1 - \delta_{st}). \end{aligned} \quad (\text{B6})$$

Unfortunately, h_{st} is not invertible, since

$$\sum_t h_{st} \lambda_t = 0; \quad (\text{B7})$$

so f in Eq. (B4) cannot be h^{-1} . This difficulty can be remedied in the following manner. Define

$$\begin{aligned} h'_{st} &= h_{st} + \alpha \delta_{s0} \delta_{t0}, \\ f'_{st} &= (h'^{-1})_{st}. \end{aligned} \quad (\text{B8})$$

Then it is not hard to show (at least for $k=2$) that when f is replaced by f' in Eq. (B4), $c_{rs,tu}$ is independent of α and remains a solution of the matrix equation that Eq. (B4) was supposed to solve. We can then calculate $c_{rs,tu}$ from Eqs. (B4) and (B8). The order of magnitude of these coefficients is

$$\begin{aligned} c_{01,01} &\sim c_{02,02} \sim c_{01,02} \sim \lambda^2/r^2, \\ c_{01,12} &\sim c_{02,12} \sim \lambda/r^3, \\ c_{12,12} &\sim 1/r^2 \end{aligned} \quad (\text{B9})$$

with the symmetry relations $c_{rs,tu} = -c_{rs,ut}$.

The calculation of the one-loop action now goes through exactly as in Ref. 10 and we arrive at Eq. (4.4) with Π replaced by Π^{JNR} . The dependence of Γ on any of the collective coordinates γ_n is given in both cases by

$$\begin{aligned} \frac{\partial \Gamma}{\partial \gamma_n} &= - \frac{1}{4\pi^2} \int d^4x \frac{\partial A_\mu^a(x)}{\partial \gamma_n} \frac{1}{\Pi(x)^2} \\ &\times \sum_{rstu} c_{rs,tu} \phi_{rsb}^\dagger(x) \epsilon_{bac} D_{\mu cd} \phi_{tud}^\dagger(x). \end{aligned} \quad (\text{B10})$$

This is a convergent integral. (In the JNR case, this is apparent only after using the fact that the leading behavior of A_μ^a for large x is independent of all the parameters γ_n .) We can therefore use Eqs. (B9), (B3), and the methods of Sec. II to estimate its size. The result is

$$\frac{\partial \Gamma}{\partial \gamma_n} \lesssim \frac{\lambda^3}{r^4}, \quad (\text{B11})$$

which implies, for the case where γ_n is a scale

$$\Gamma \lesssim \frac{\lambda^4}{r^4}. \quad (\text{B12})$$

Thus Γ is less than the terms of interest in Sec. IV and may be neglected there.

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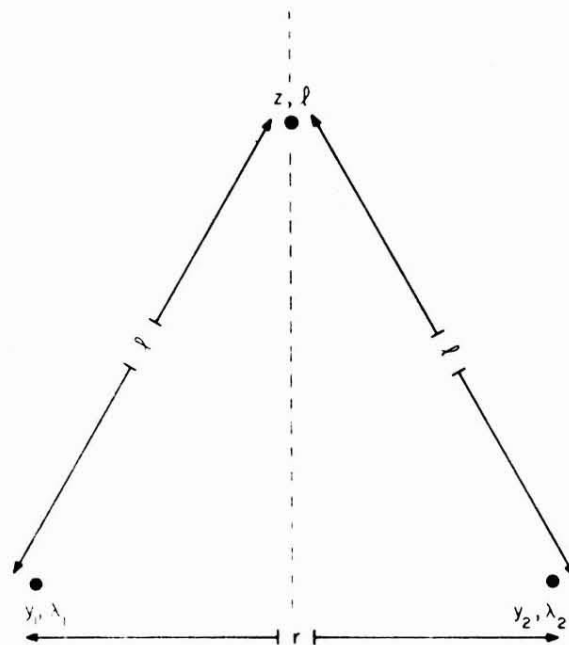


FIG. 1. The location of the points and the corresponding scales of the JNR $k=2$ solution in the widely separated limit ($\lambda_1, \lambda_2 \ll r$). We have used the freedom inherent in the solution to choose z on the hyperplane (dashed line) which perpendicularly bisects the line between y_1 and y_2 , and to choose the scale l corresponding to the point z such that $l = |y_1 - z| = |y_2 - z|$.